



Embryology of the genitourinary system

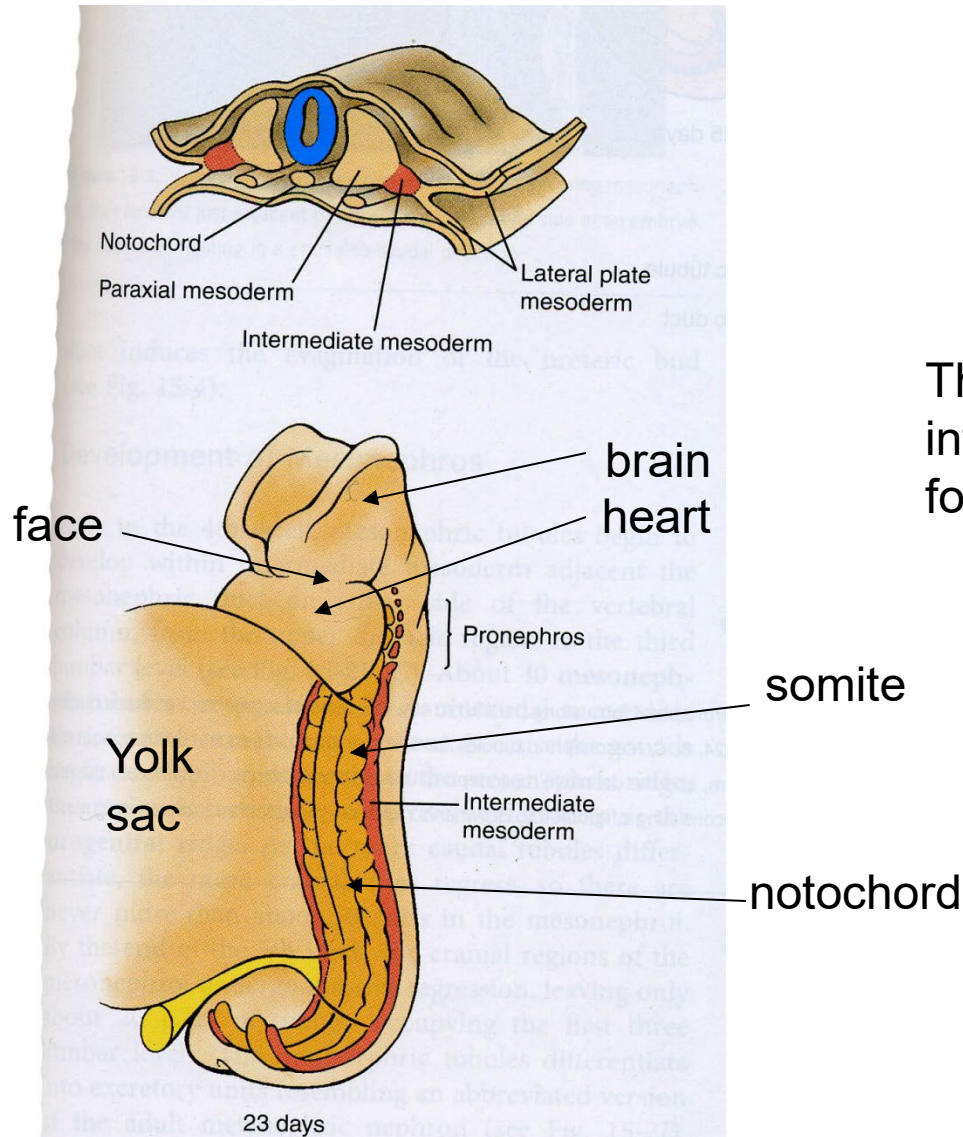
Medical Gross Anatomy

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Professor

Session Objectives

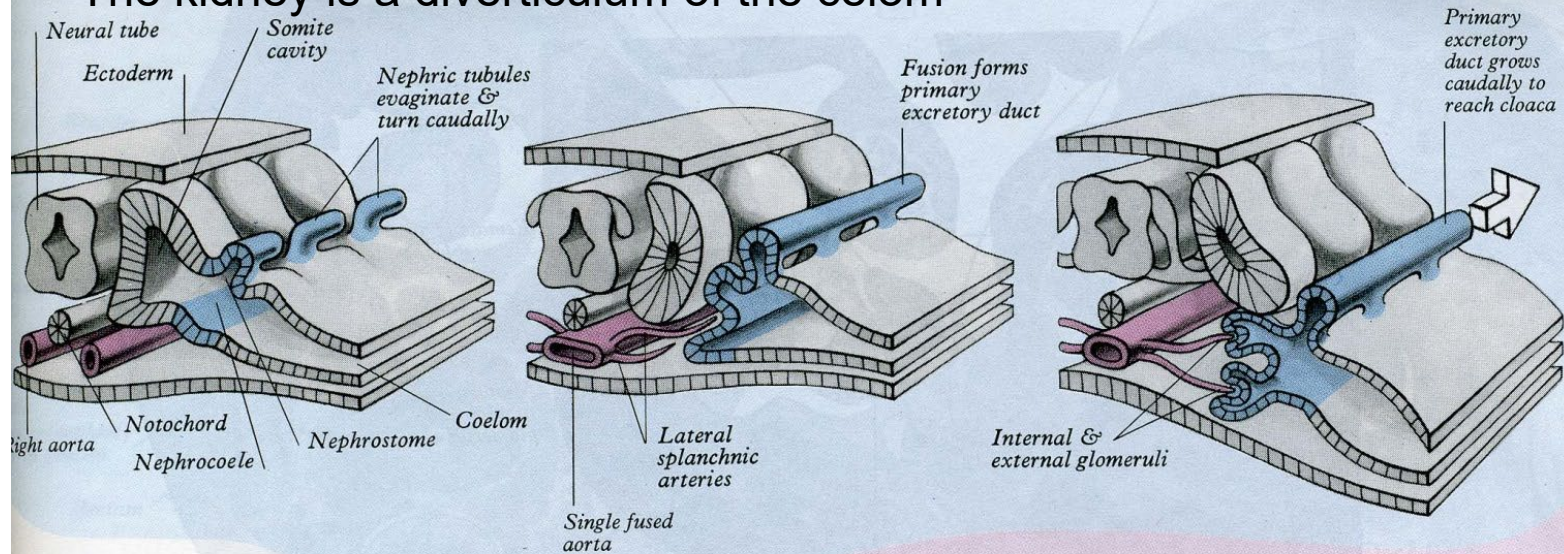
1. embryology of the intermediate mesoderm and mesonephric duct systems
2. glomerulus
3. ureteric bud and metanephric blastema
4. embryologic derivatives of the cloaca and urogenital sinus
5. selected pathologies and embryologic anomalies provided



The intermediate mesoderm forms the kidneys

Figure 15-1. The intermediate mesoderm gives rise to paired, segmentally organized buds from the cervical to the sacral region. The pronephros is initially formed early in the 4th week in the cervical region.

The kidney is a diverticulum of the celom



NEPHROGENIC CORD OF INTERMEDIATE MESODERM

PRONEPHROS

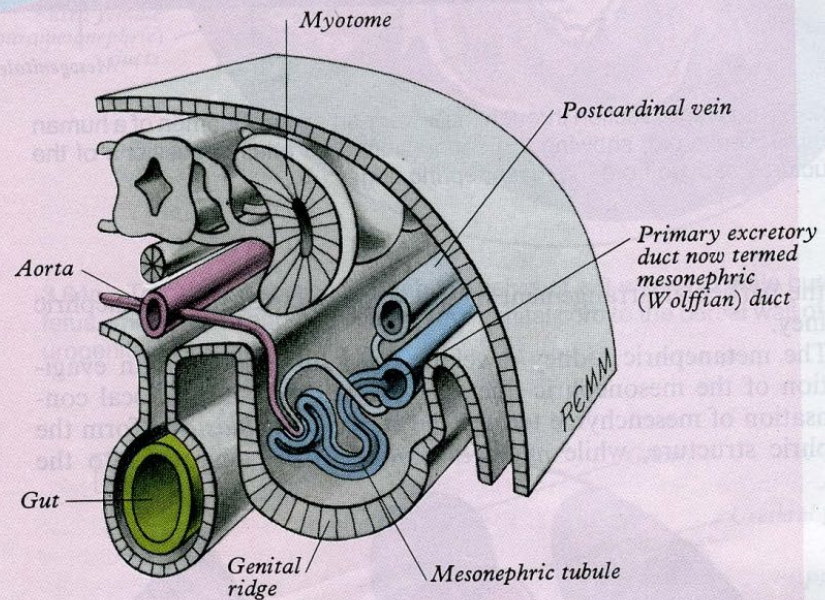
Rudimentary in mammals

MESONEPHROS

METANEPHROS

Mesonephric duct

Mesonephric tubules



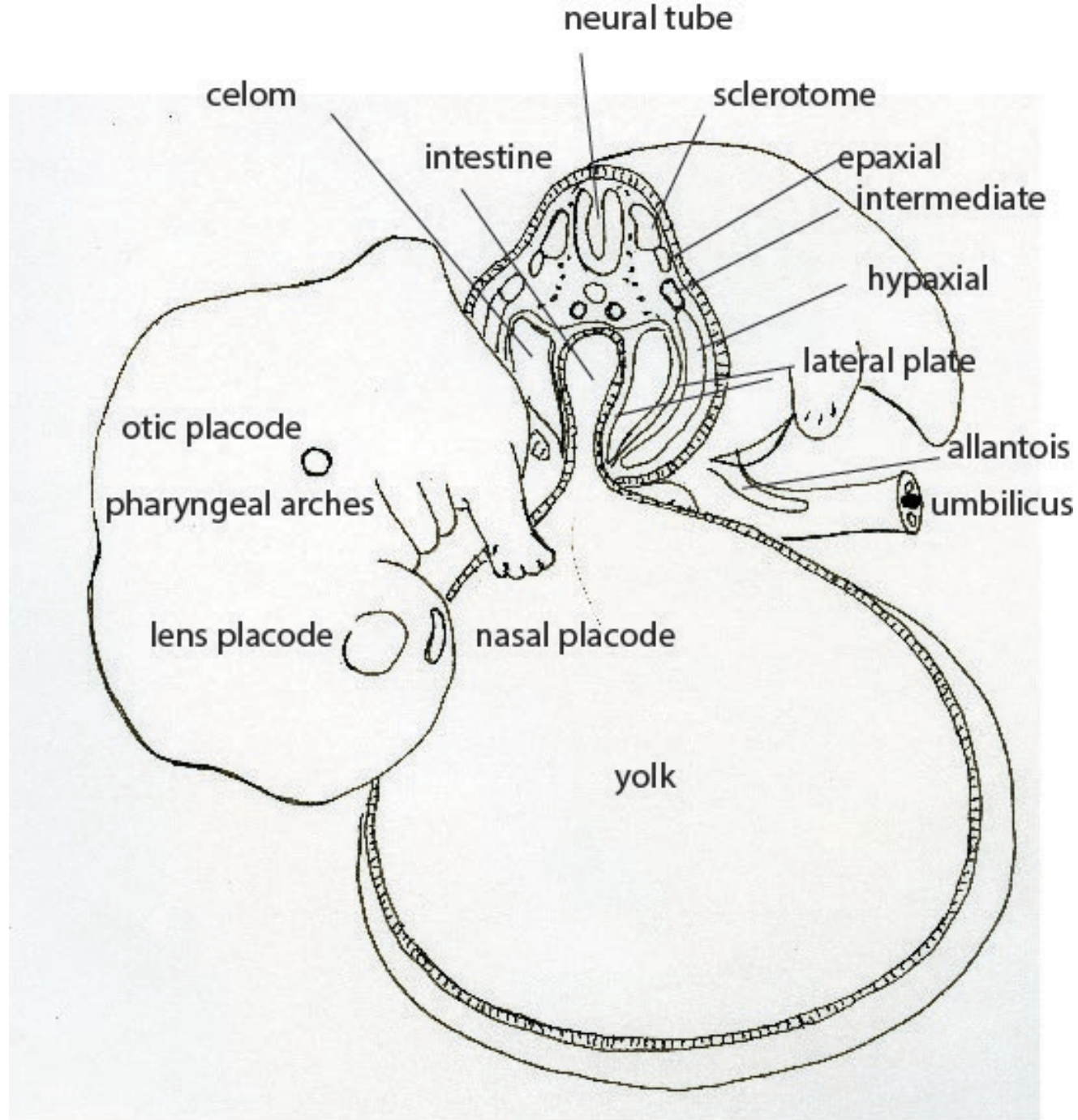
All nephric structures, derive from the intermediate mesoderm

Three types of kidneys found in embryos and also representing evolutionary stages and found in fishes

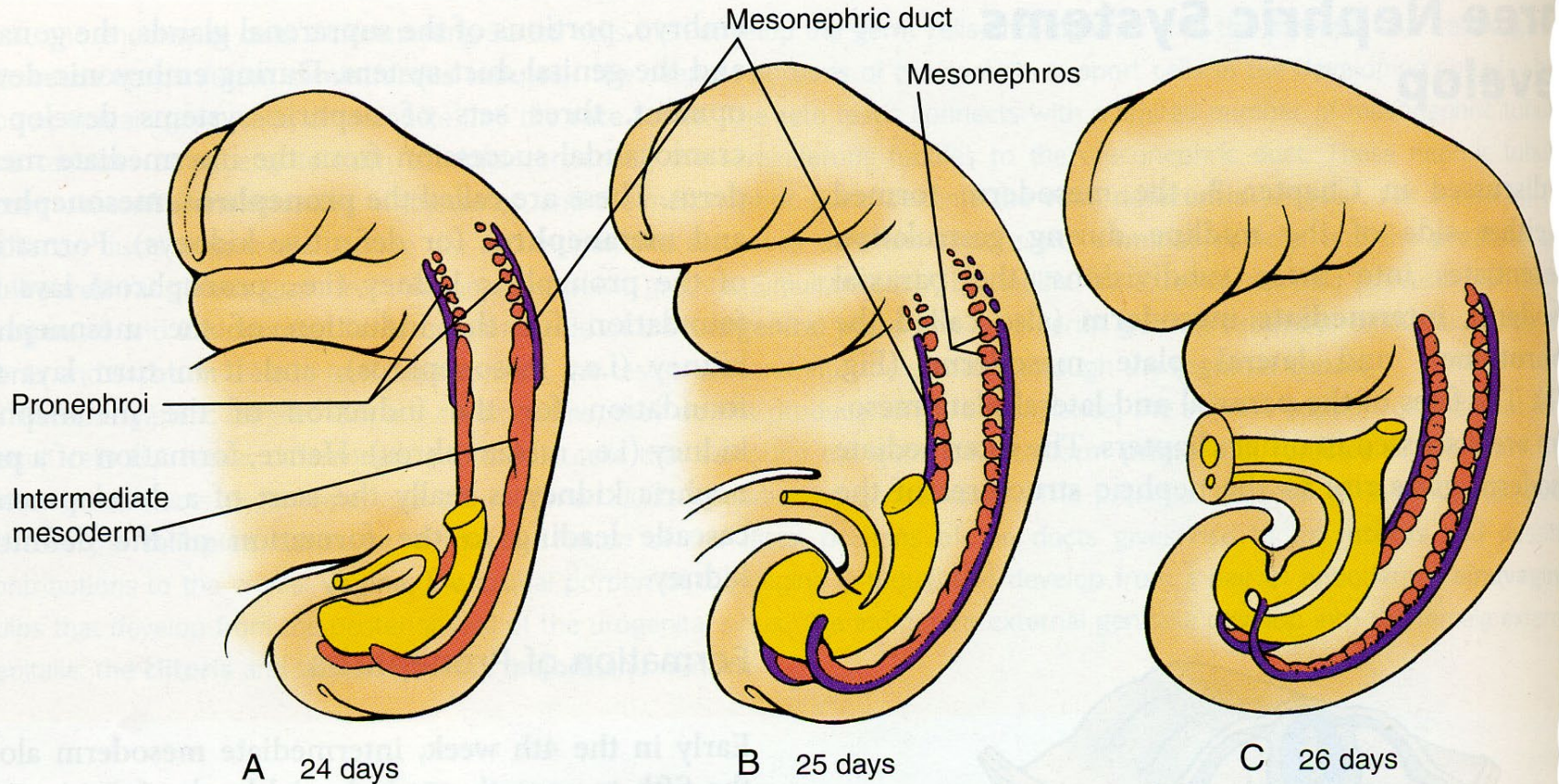
Pronephros is very primitive, and never develops into functional excretory structures in the human.

Mesonephros – its ducts and tubules remain in the adult – the glomeruli disappear in the adult.

Metanephros is the adult kidney with new glomeruli.



the intestine is a diverticulum of the yolk sac



The mesonephros and the mesonephric kidney develops. This kidney disintegrates in part as the metanephric kidney forms. In part, as the mesonephric tubes remain.

Originally, in fishes the mesonephric kidney excreted urine and eggs and sperm. Later the mesonephric kidney got involved with male reproduction and the metanephric kidney took the role of urine excretion.

Wolffian duct = mesonephric duct

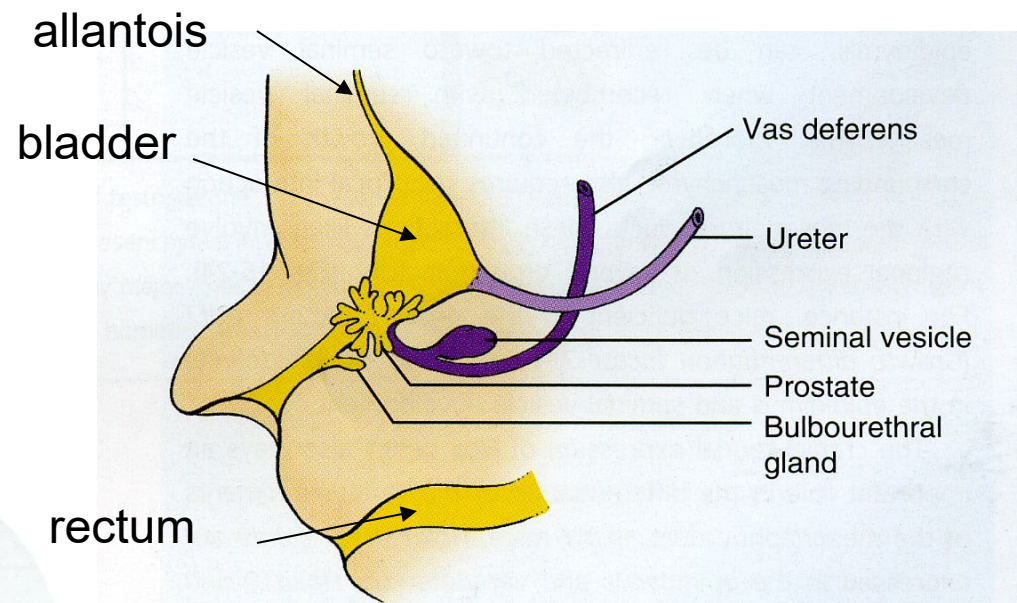
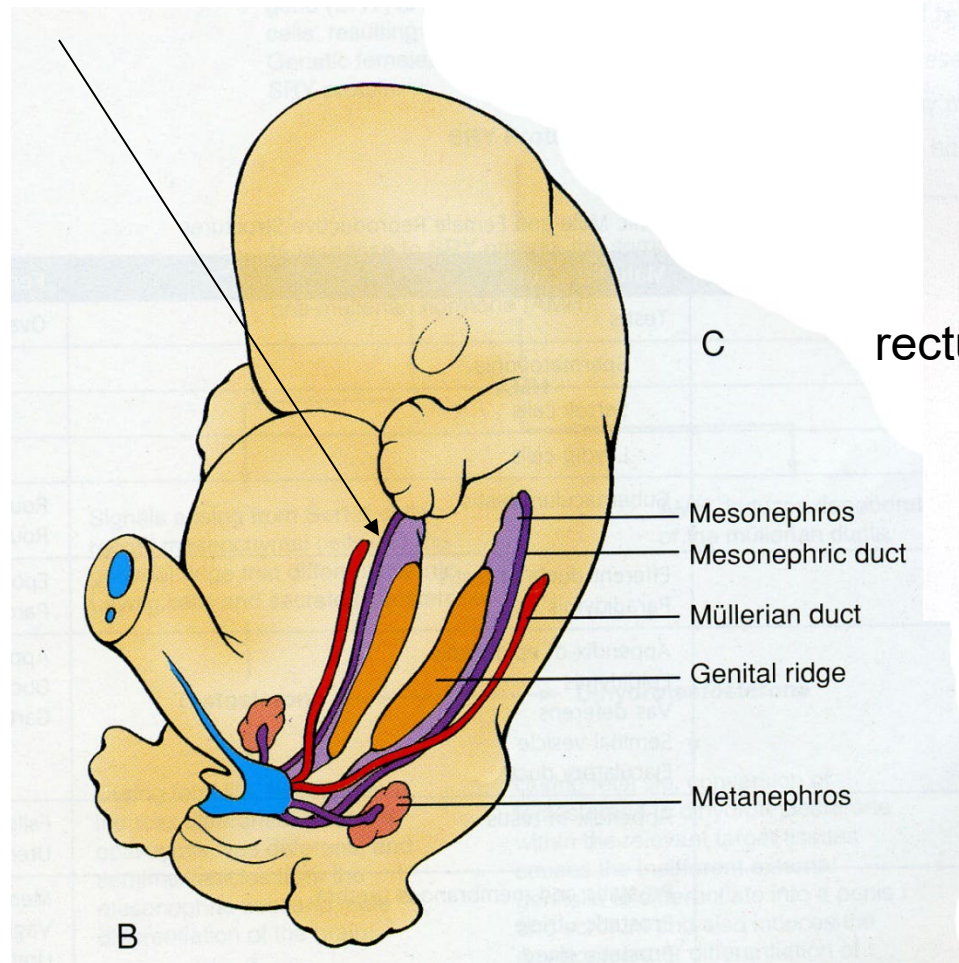


Figure 15-23. Development of the seminal vesicles, prostate, and bulbourethral glands. These glands are induced by androgens between the 10th and 12th weeks.

The colors in this slide are backwards

Mesonephric duct or Wolffian duct --- Müllerian duct = paramesonephric duct

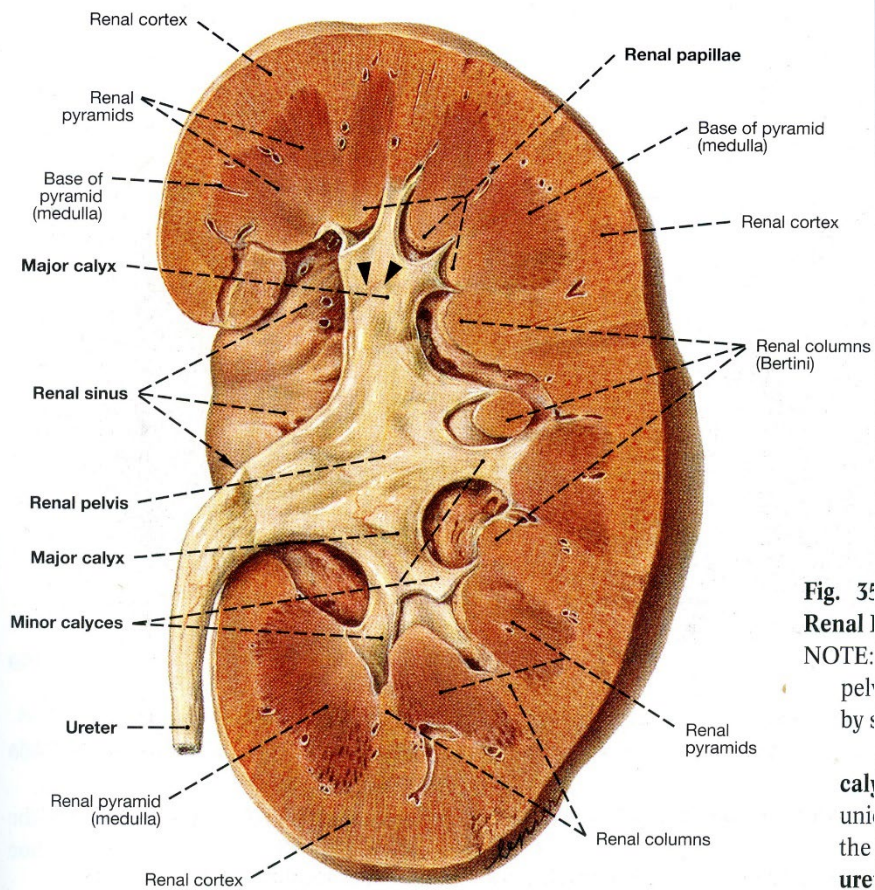
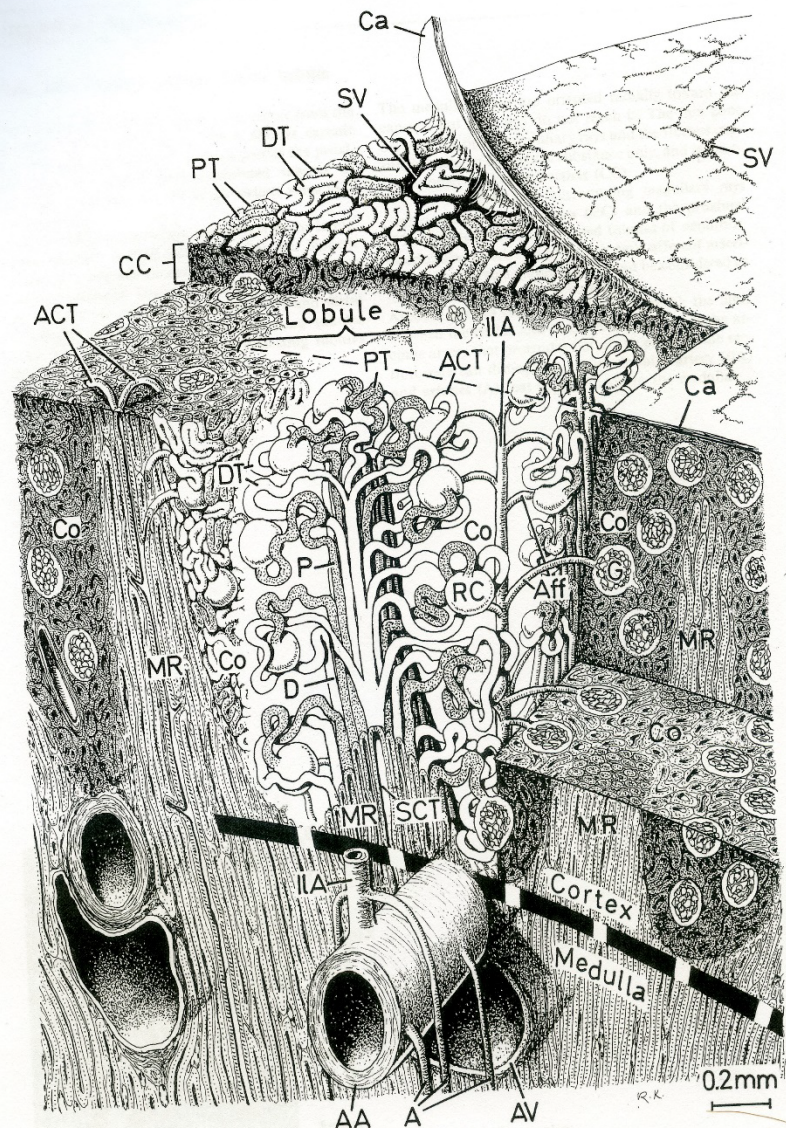
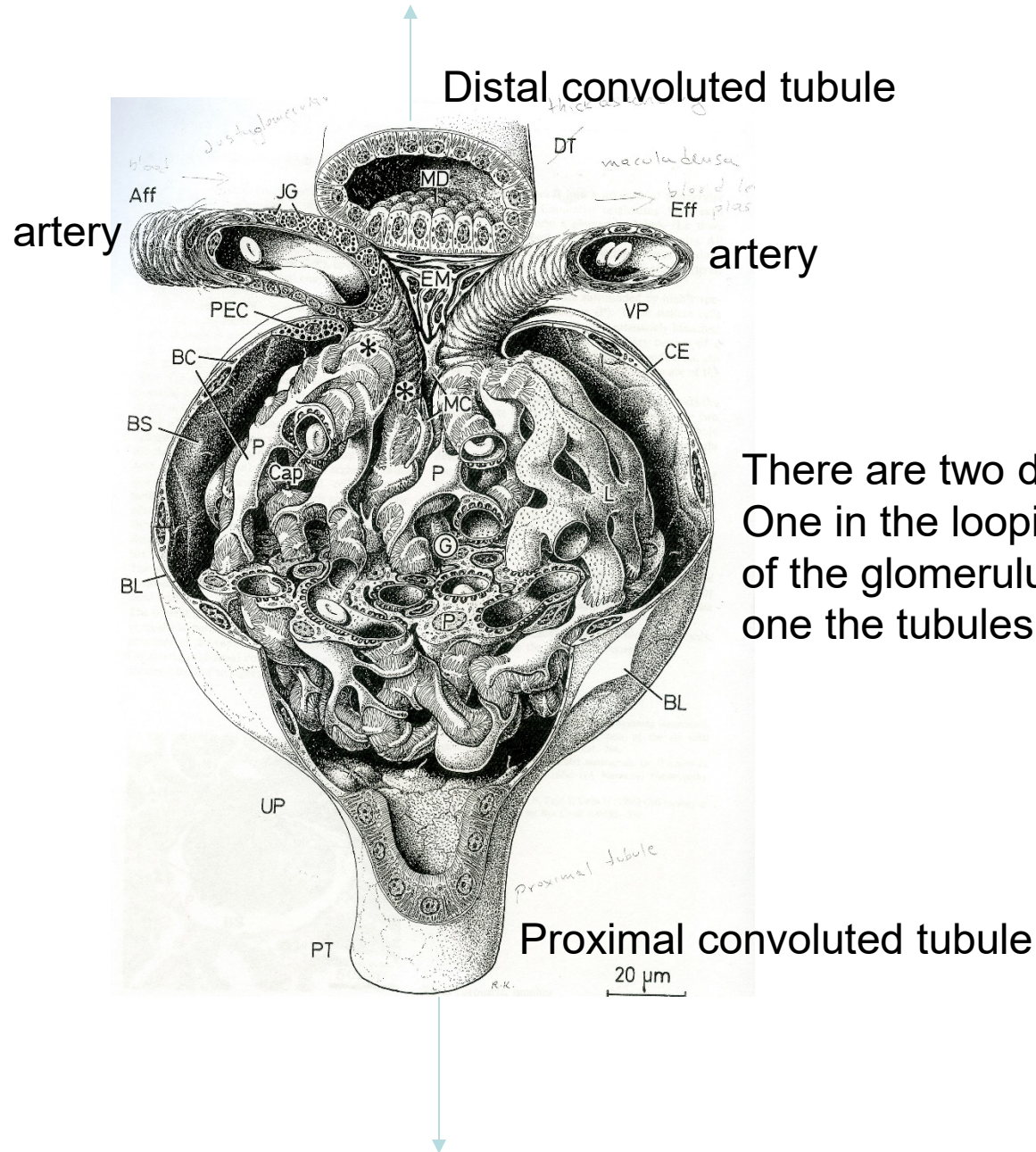


Fig. 32
Renal
NOTE:
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A glomerulus



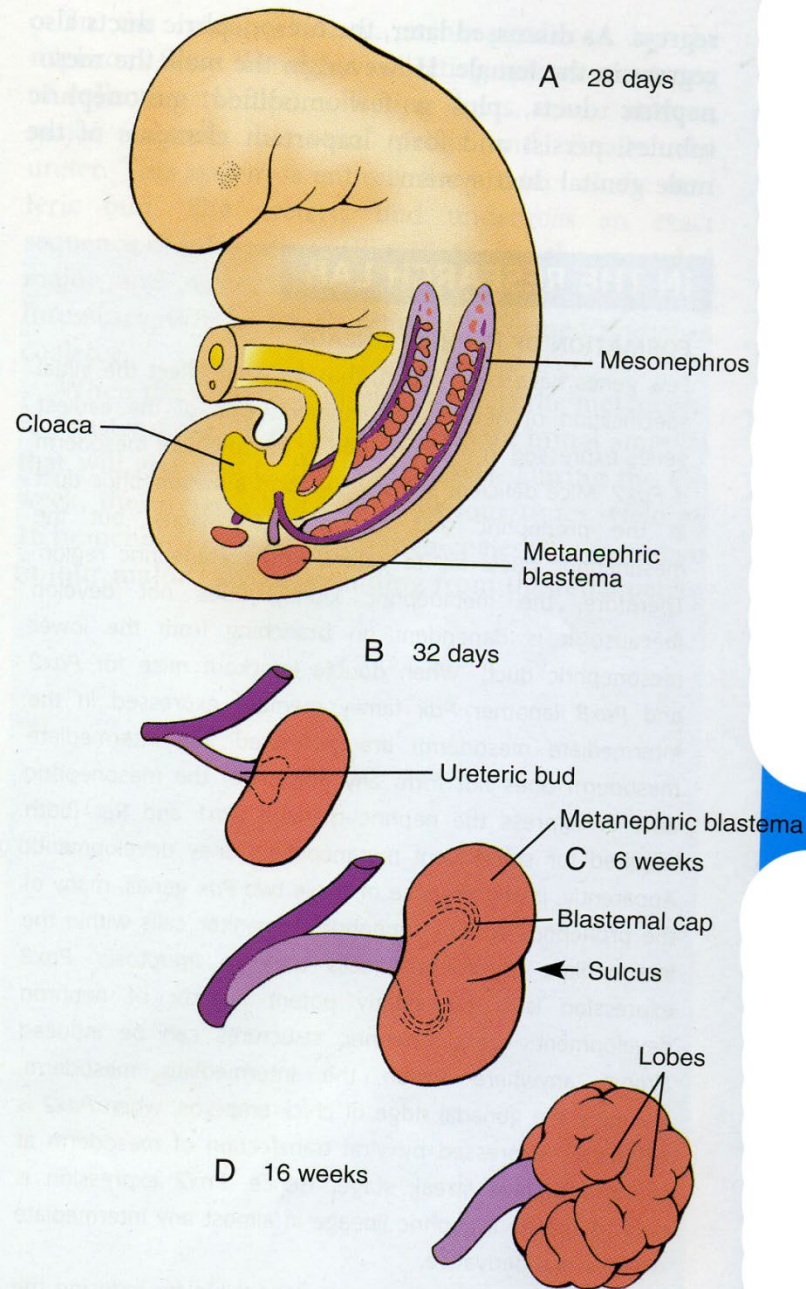
Mesonephric - Ureteric bud forms the **collecting portion** of the kidney:
Ureter, renal pelvis, major and minor calyces, collecting ducts,
collecting tubules

Metanephric blastema sends signals to the ureteric bud causing it to
branch and grow, and the ureteric bud induces the blastema to
condense and form tubules INTERACTION

Metanephric blastema forms **excretory portion**: Bowman's capsule,
proximal convoluted tubule, loop of Henle, distal convoluted tubule

Metanephric kidney- ureteric buds branch from
the mesonephric ducts, and they interact with a mass of intermediate
mesoderm called the metanephric blastema
(also referred to as metanephric mesenchyme)

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Ureteric bud as a diverticulum of the mesonephric duct

Figure 15-4. Origin of the metanephric kidneys. A, A metanephric blastema develops from intermediate mesoderm.

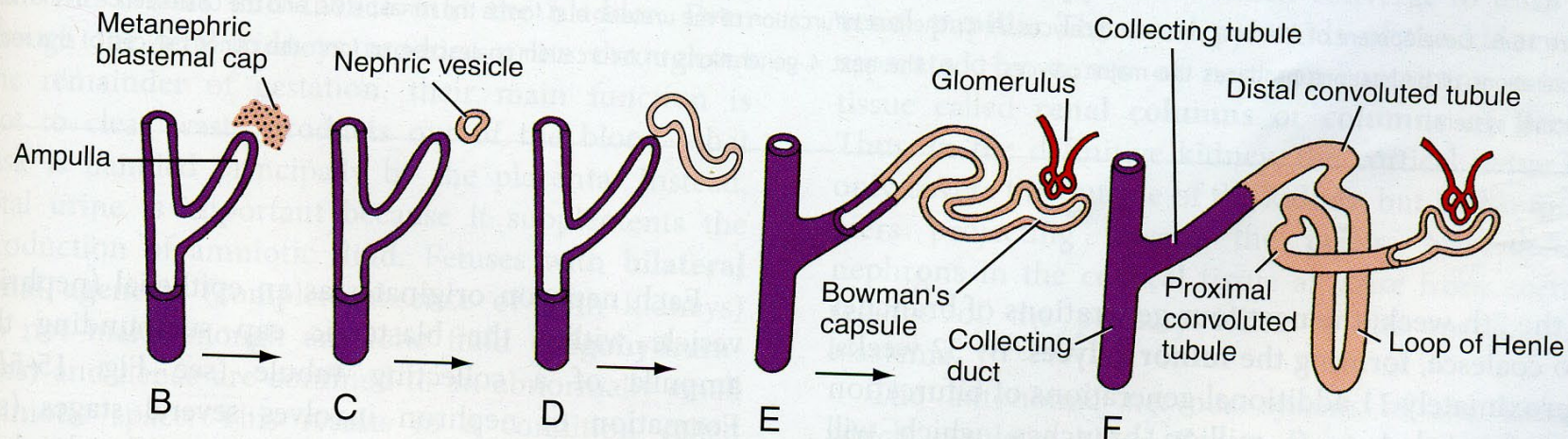


Figure 15-5. Development of the renal collecting system and nephrons. A, The ureteric buds continue to bifurcate until the 32nd week, producing 1 to 3 million collecting tubules and ducts. B-F, The tip of each collecting tubule induces the development of a metanephric blastemal cap, which differentiates into a nephric vesicle. This vesicle ultimately forms a Bowman's capsule and the proximal and distal convoluted tubules and loops of Henle. Functional nephric units (of the type shown in E) first appear in distal regions of the metanephros at 10 weeks.

Table 15-1 Structures Composing the Collecting and Excretory Portions of the Metanephric Kidney

Collecting Portion (Ureteric Bud)	Excretory Portion (or Nephron) (Metanephric Blastema)
Ureter	Bowman's capsule
Renal pelvis	Proximal convoluted tubule
Major and minor calyces	Loop of Henle
Collecting ducts	Distal convoluted tubule
Collecting tubules	

Fate of mesonephric kidney

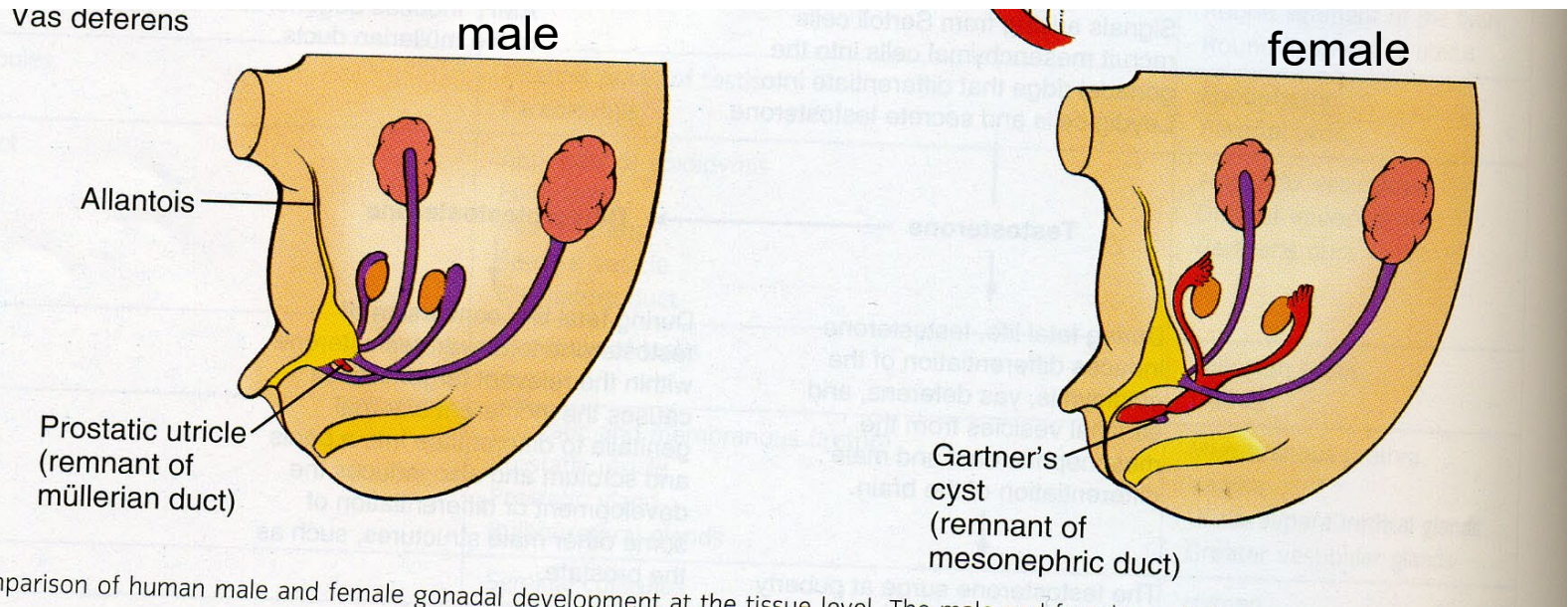
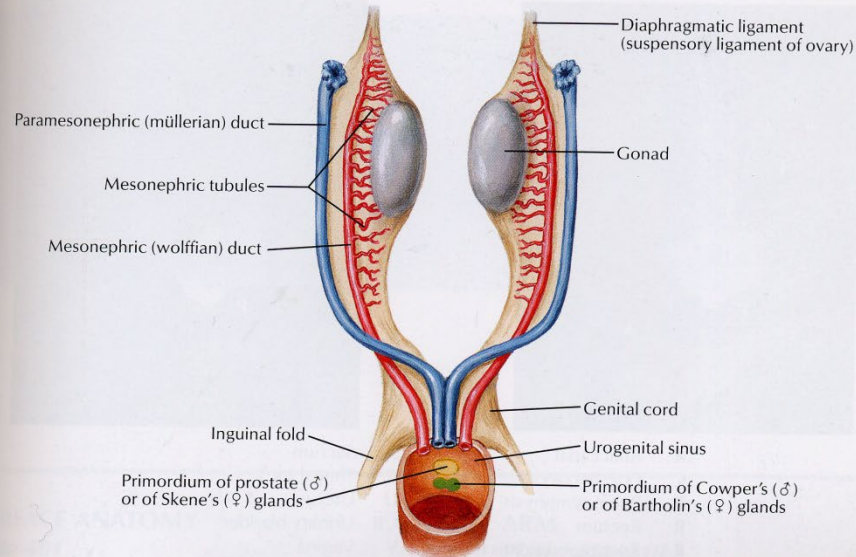


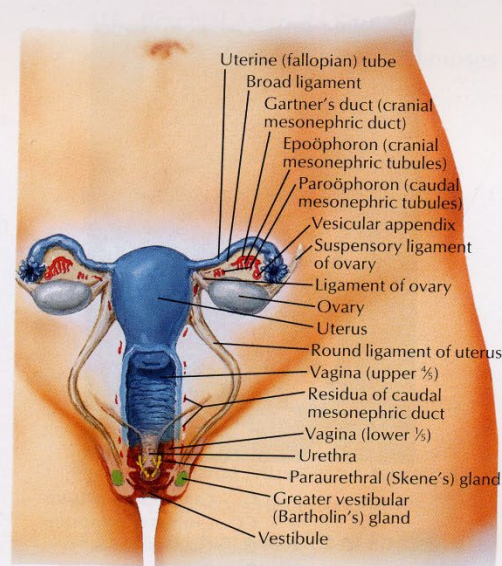
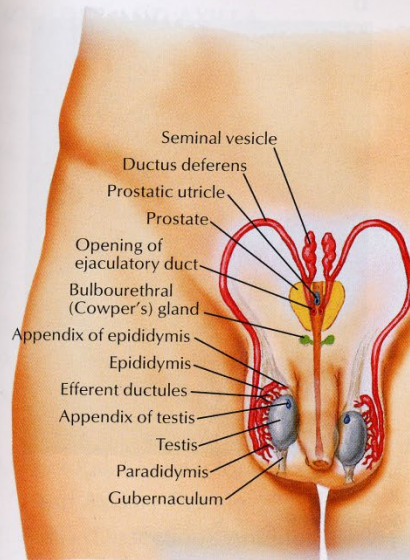
Figure 15-19. Comparison of human male and female gonadal development at the tissue level. The male and female

Mesonephric kidney is not lost in adults –
Its tubing becomes incorporated in the reproductive system
of males – vas deferens and epididymis - trigone, ureter and metanephric calyces.

In females the paramesonephric duct is derived from the uterus as a diverticulum

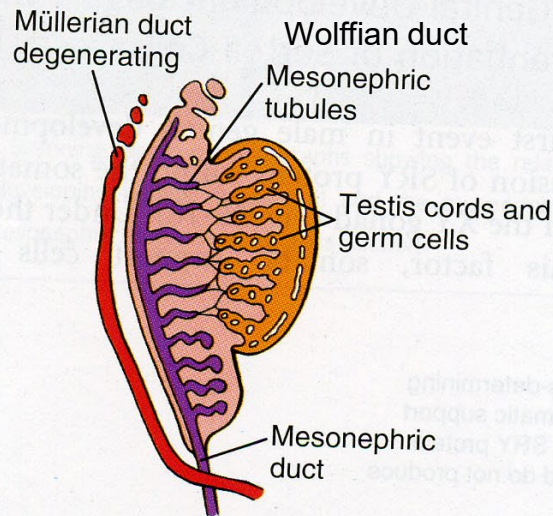


Undifferentiated
Male ← Female

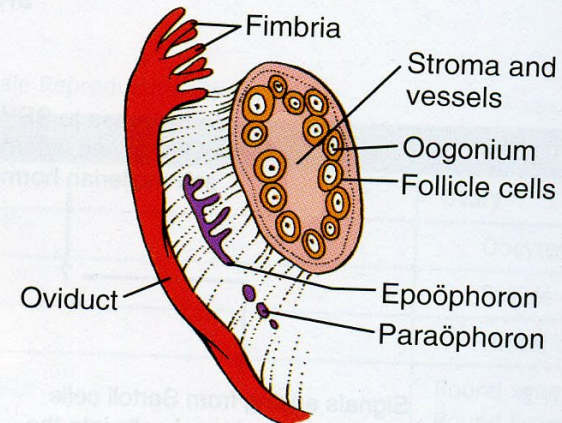
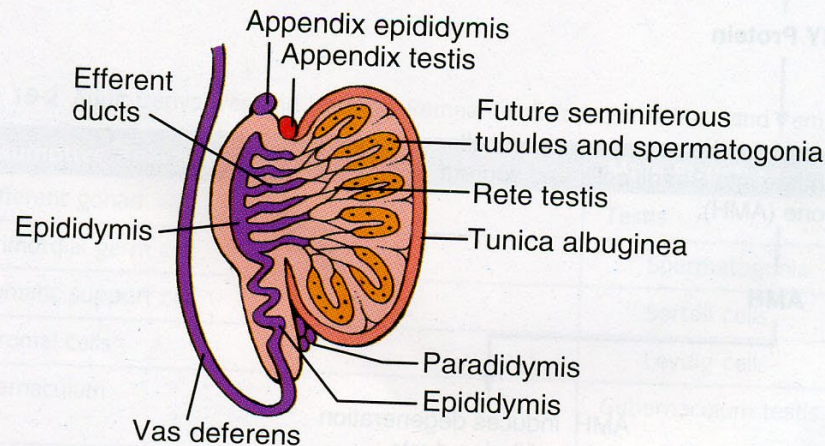
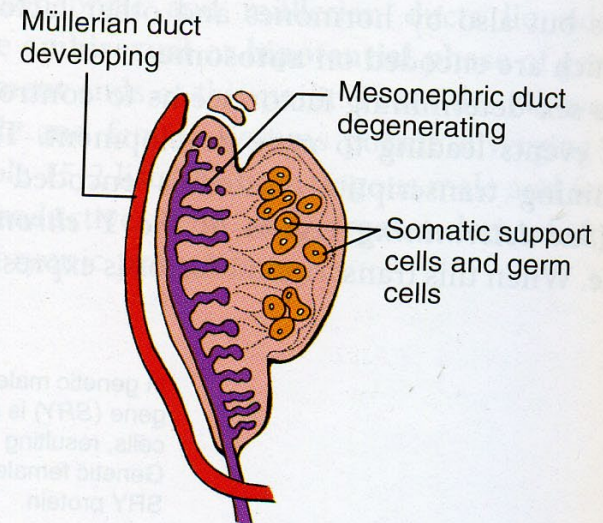


Paramesonephric duct

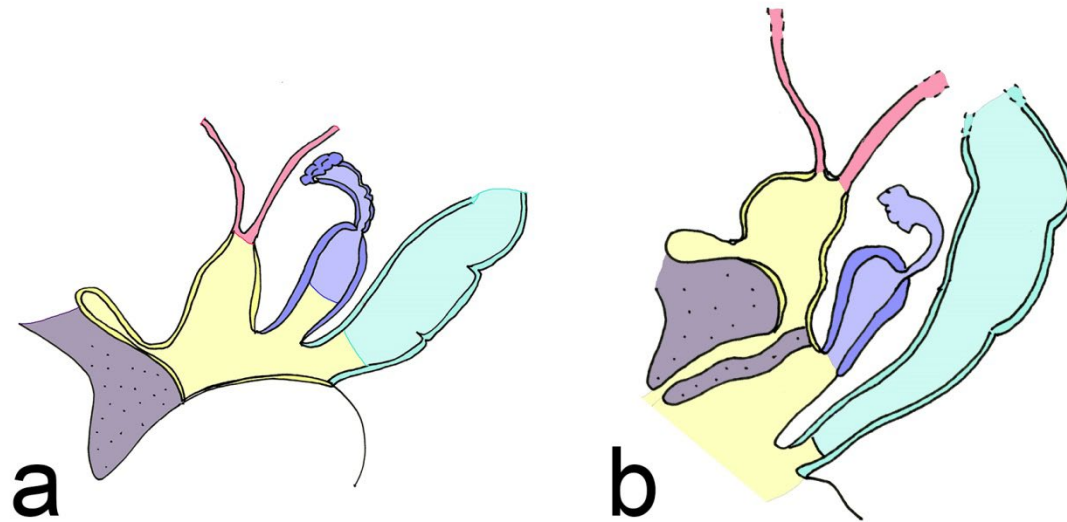
Male



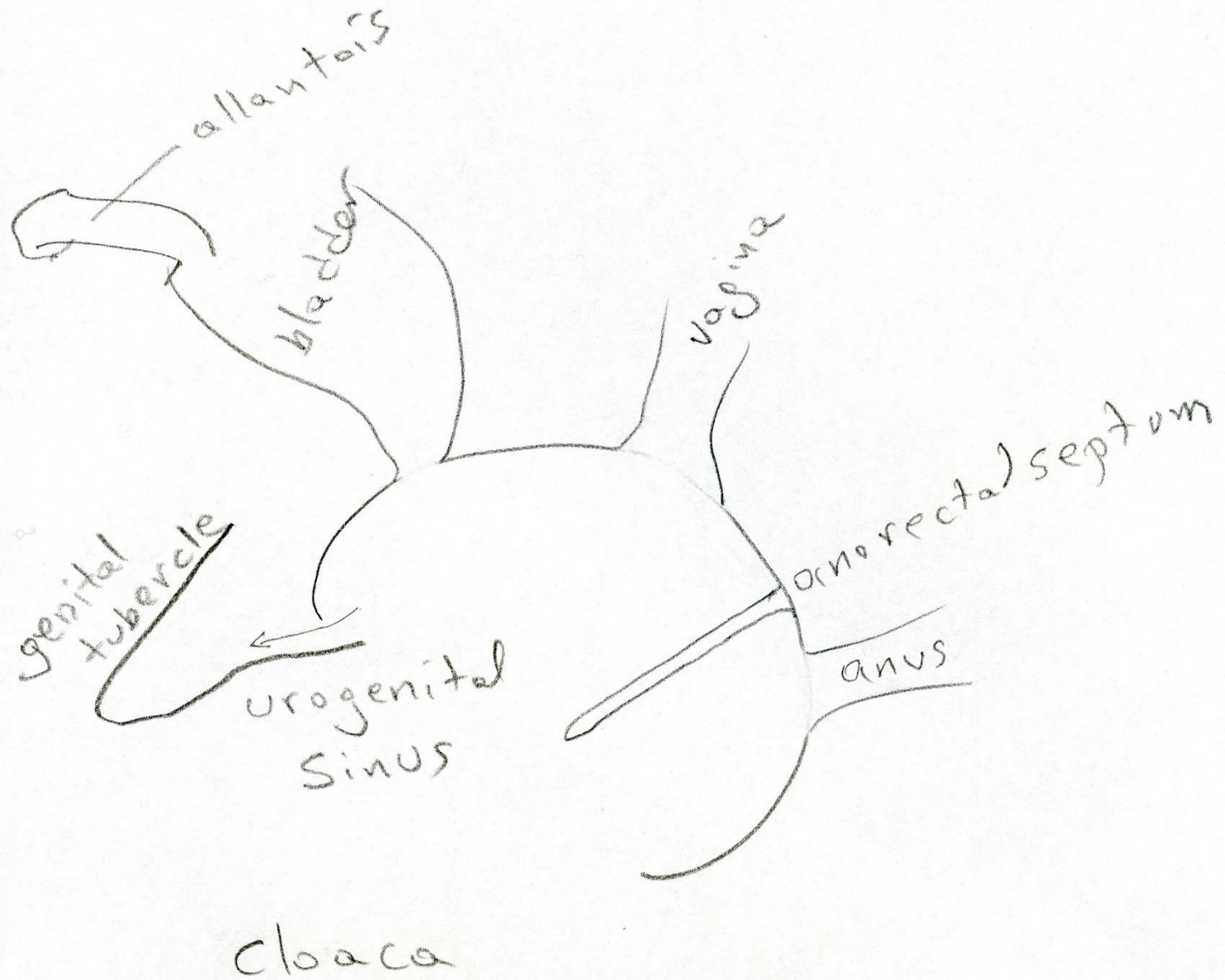
Female



The colors are backward



The cloaca and its derivatives in a female. The cloaca is shown in yellow, genital tubercle in purple, paramesonephric duct in blue, and endoderm of the gastrointestinal tract in turquoise. A- Early stage, where the distal urinary, reproductive, and gastrointestinal tracts all have a common opening. The anterior out pouching towards the genital tubercle represents the allantois, which is an early drainage system for the fetus. B- Later stage where the urogenital system is separated from the distal GI tract via the urorectal septum. From anterior (left) to posterior (right), structures include the bladder and urethra, developing uterus and vagina, and anorectal canal. Watch the video.



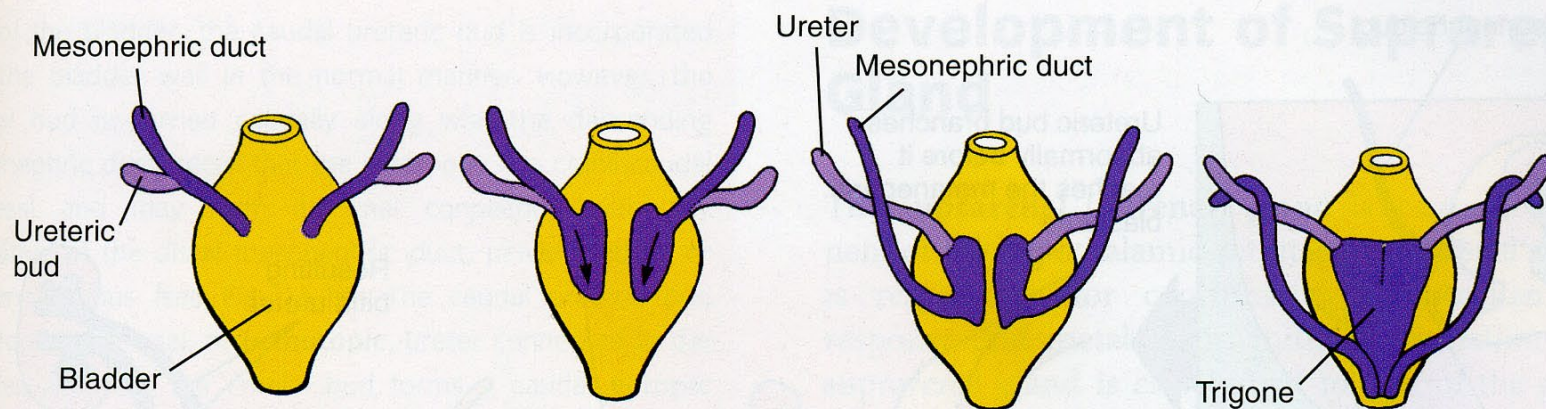


Figure 15-12. Exstrophy of the mesonephric ducts and ureters into the bladder wall. Between weeks 4 and 6, the root of the mesonephric duct exstrophies into the posterior wall of the developing bladder. This process brings the openings of the ureteric buds into the bladder wall, whereas the opening of the mesonephric duct is carried inferiorly to the level of the pelvic urethra. The triangular region of exstrophied mesonephric duct incorporated into the posterior bladder wall forms the trigone of the bladder.

Exstrophy of mesonephric ducts is the formation of trigone

The distal portion of the mesonephric ducts become incorporated into the posterior wall of the cloaca (process is called extrophy)- this forms the trigone of the bladder

Ureteric bud	Ureter, renal pelvis, major/minor calyces, collecting ducts, collecting tubules
Metanephric blastema	Bowman's capsule, proximal convoluted tubule, loop of henle, distal convoluted tubule
Mesonephric duct	Trigone of the bladder (and gives rise to the ureteric bud)
Urogenital sinus (part of the cloaca)	Bladder, urethra

Vascularization

Metanephric kidneys form low near the bladder and slowly ascend to the adult position

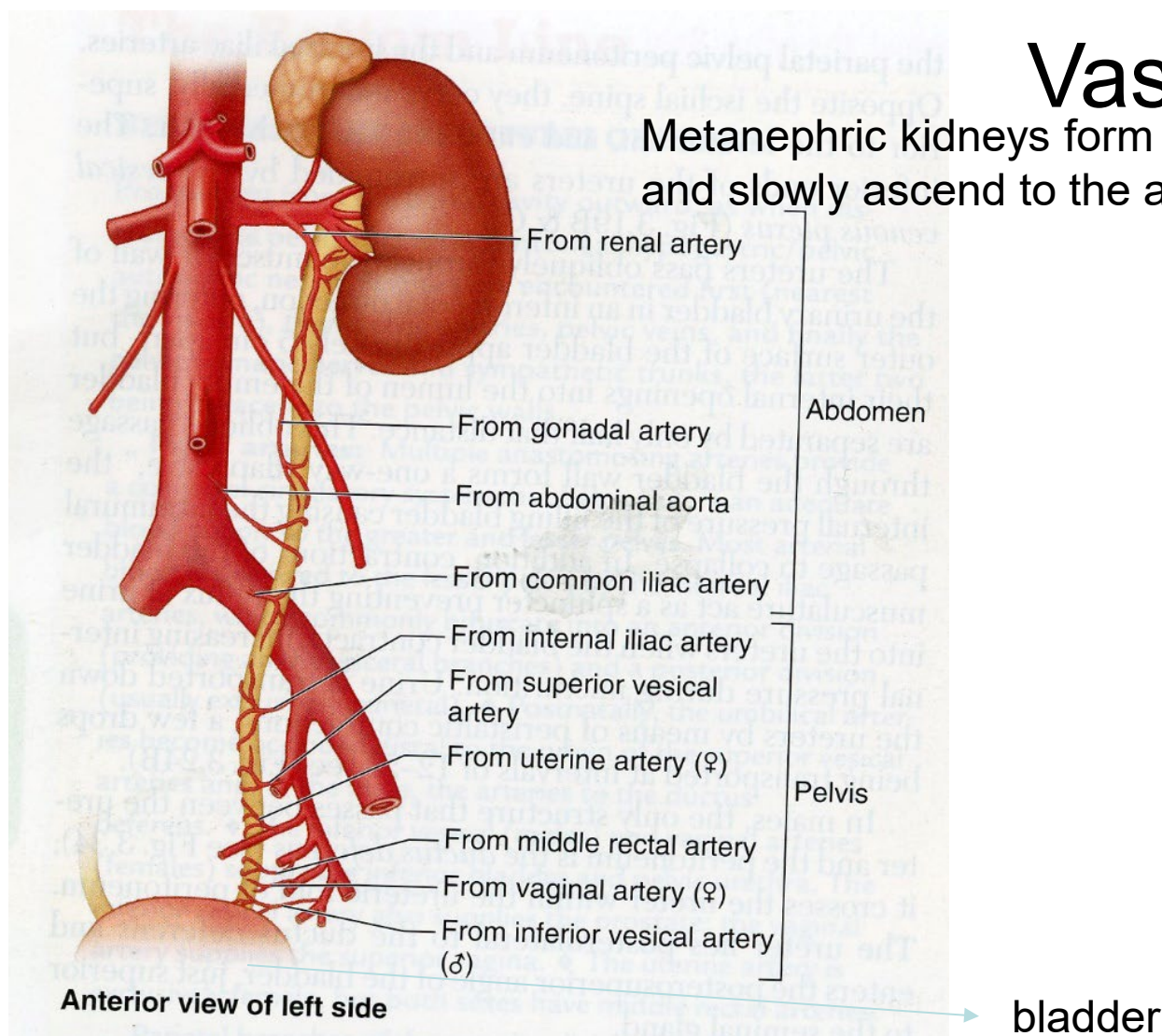


FIGURE 3.25. Blood supply of ureter. Branches supplying the abdominal half of the ureter approach medially, while those supplying the pelvic half approach laterally. During surgery, the ureters are avoided and left undisturbed when possible. If necessary, traction of the ureters is applied gently and only toward the blood supply to avoid disruption of the small branches.

Why this system is called urogenital? Where the two systems meet

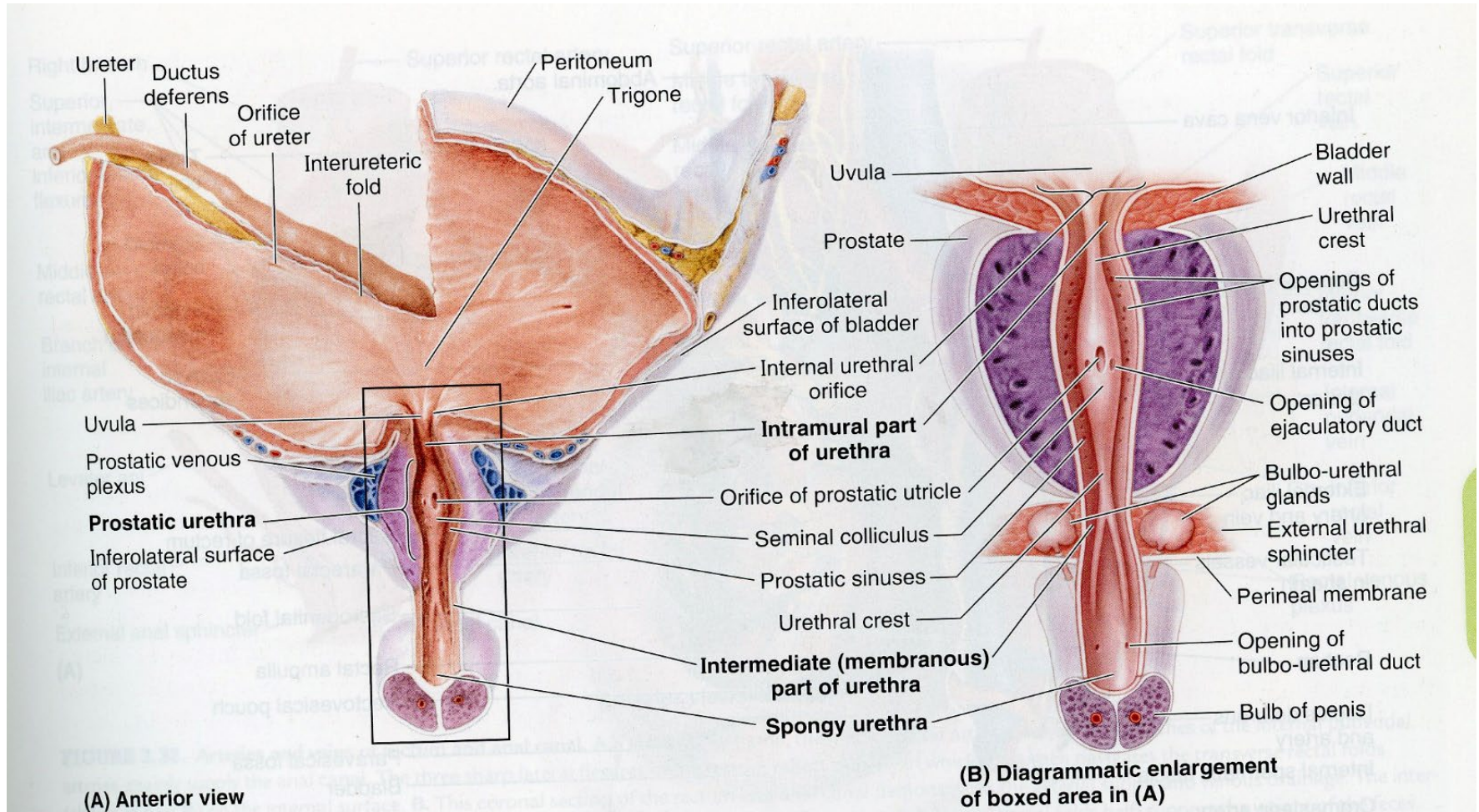


FIGURE 3.30. Interior of male bladder and urethra. **A.** The anterior parts of the bladder, prostate, and urethra are cut away. A portion of the posterior wall of the bladder has been removed to reveal the intramural part of ureter and the ductus deferens posterior to the bladder. The interureteric fold runs between the entrances of the ureters into the bladder lumen, demarcating the superior limit of the trigone of the bladder. The prominence of the posterior wall of the internal urethral orifice (at the tip of the leader line indicating this orifice), when exaggerated, becomes the uvula of the bladder. This small projection is produced by the middle lobe of the prostate. The prostatic fascia encloses the prostatic venous plexus. **B.** This enlarged detail of the boxed area in **A** demonstrates the bulbo-urethral glands embedded in the substance of the external urethral sphincter.

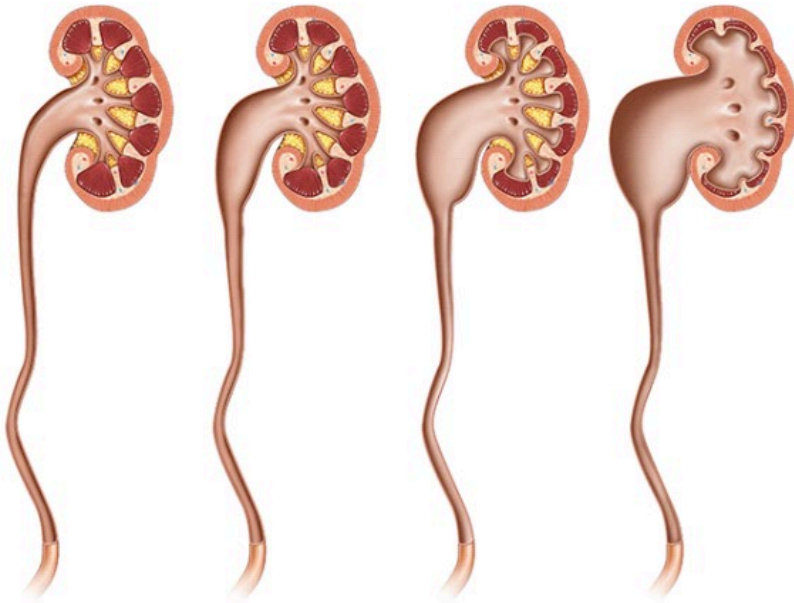
Common pathologies

Imperforate anus



- Dysfunction of urorectal septum
- Rectum opens into urinary or reproductive tract, or ends in a blind pouch
- Clinical: failure to pass meconium
- VACTERL association:
(vertebral anomalies, anal atresia, cardiovascular abnormalities, tracheoesophageal fistula, esophageal atresia, renal anomalies, limb defects)

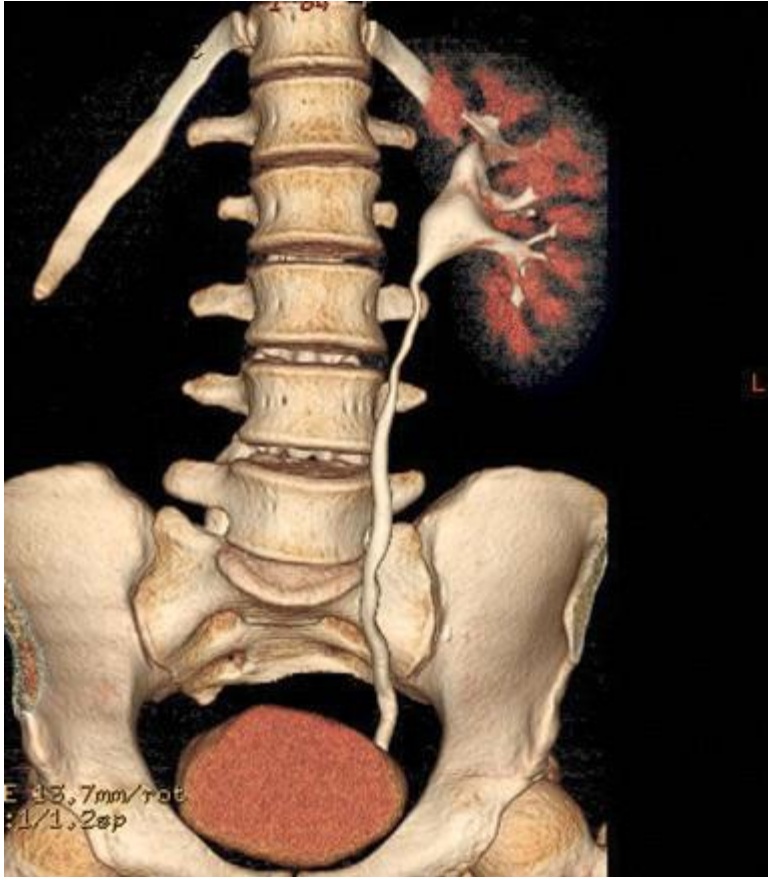
Hydronephrosis



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- Fluid filled enlargement of one (or both) kidneys
- In the fetus, this is likely due to obstruction in the ureters
- Most common site for obstruction: ureteropelvic junction

Renal Agenesis



- Ureteric bud fails to contact the metanephric blastema → kidney does not form
- Unilateral: remaining kidney undergoes hypertrophy
- Bilateral: severe oligohydramnios, incompatible with life

Oligohydramnios

- Insufficient amounts of amniotic fluid surrounding the fetus
- Potter sequence: deformed limbs, dry skin, abnormal faces
- Pulmonary hypoplasia

Potter Sequence



Polyhydramnios



- Excess accumulation of amniotic fluid
- Caused by: lack of fetal swallowing (GI malformations, anencephaly), maternal diabetes mellitus

OLIGOHYDRAMNIOS

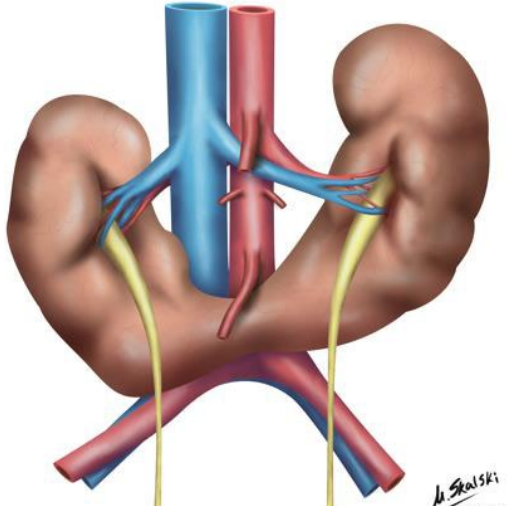
Low amniotic fluid levels

Oligohydramnios occurs when the volume of fluid in the amniotic sac is lower than average, usually because the placenta is not functioning properly.

When the placenta isn't working as expected, the baby gets less oxygen and nutrients. This can cause HIE, premature birth, IUGR, and other serious complications and injuries.



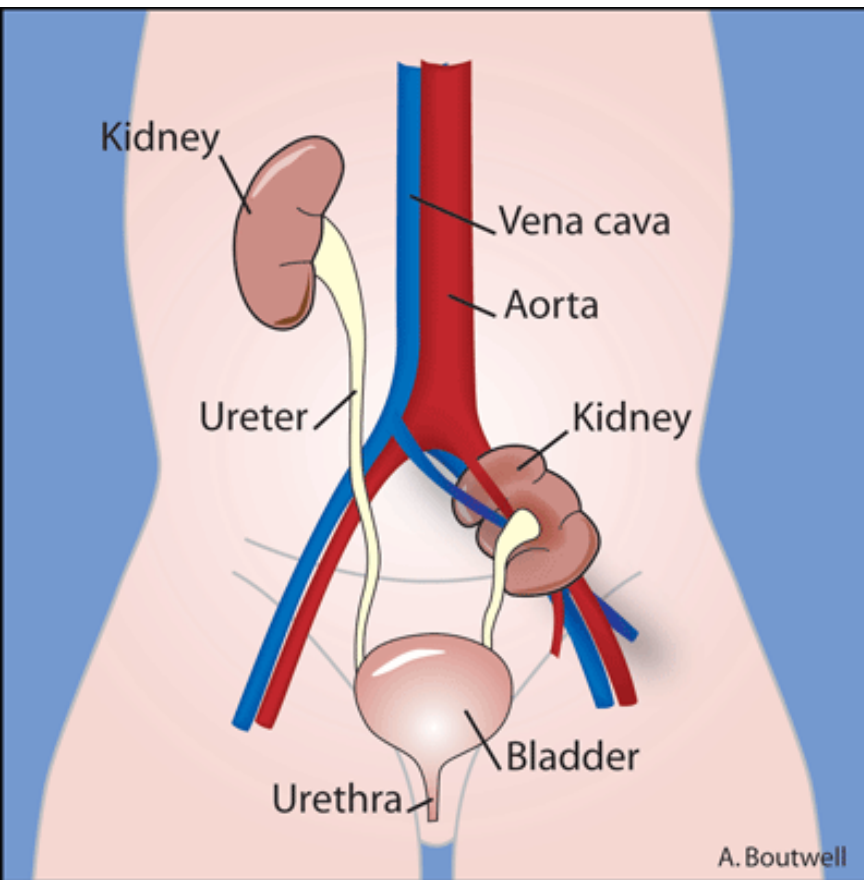
Horseshoe Kidney



- Fusion of the right and left metanephroi near midline
- Unified kidney is prevented from reaching the abdominal location by the inferior mesenteric vein
- Increased risk for UTIs and kidney stones



Pelvic Kidney



Failure of one kidney to ascend to its normal location

- Functions normally

Summary Slide

Ureteric bud	Ureter, renal pelvis, major/minor calyces, collecting ducts, collecting tubules
Metanephric blastema	Bowman's capsule, proximal convoluted tubule, loop of henle, distal convoluted tubule
Mesonephric duct	Trigone of the bladder (and gives rise to the ureteric bud) and vas deferens
Urogenital sinus (anterior part of the cloaca)	Bladder, pelvic urethra

-
- Oligo - means less
 - Poli – means much
 - Meso - means middle
 - Exstrophy - means inside out
 - Trigone – triangle
 - Agenesis – not growth – no development
 - Cloaca - Roman water clay tubes
 - Glomerulus – glomus ball of yarn (ulus is small)
 - Meta – after
 - Nephros - kidney

- <https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>